GS EXTERIOR EXPERTS · DECK DIVISION · MATERIAL TAKE-OFF + BUILD SET · FOR DECKS & DOCKS

5549 S Lansing Ct · Elevated Deck

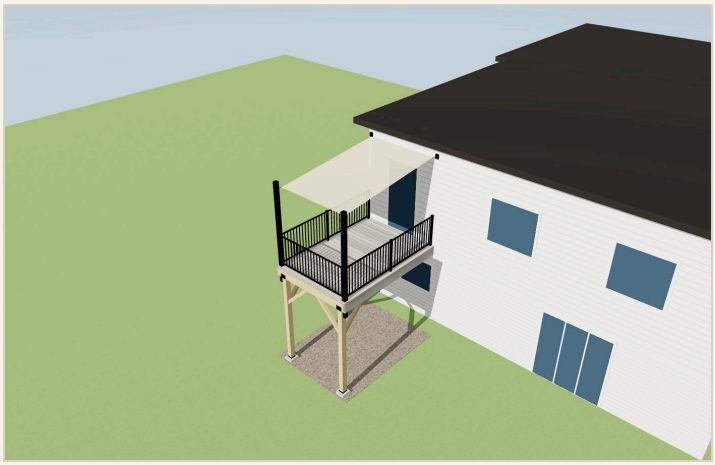
5549 S Lansing Ct, Englewood · Rachel Chiu · 8' x 12' deck, 10' above the yard, off the existing door · TrueFrame 2x10 @ 16" on a (2)4x10 Douglas fir beam and (2) 6x6 DF #1 posts on Diamond Piers · TimberTech Reserve Driftwood, single picture frame, CONCEALoc · TimberTech Fulton rail, Black · posts + anchors for the owner-supplied square shade sail · October 19, 2026

Account GSEXT1 · ship to 5549 S Lansing Ct, Englewood, CO 80111 · please quote every line; call Jade Pender 303-550-9558 with substitutions.

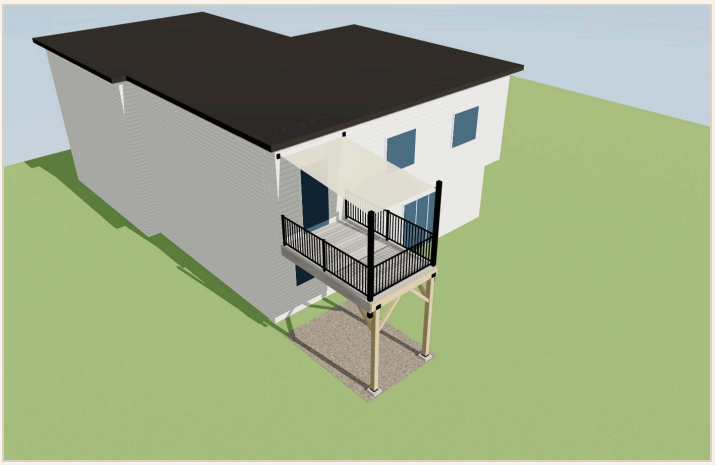
SQUARE FEET	FOOTINGS	ORDER LINES	HEIGHT
92	**2**	**36**	**10'-0"**

THE JOB

Deck	92 SF · 10'-0" above grade · 7'-9" x 11'-5 1/2" outside to outside
Joists	7 2x10 #1 True Frame Joist SYP GC @ 16" OC x 11'-1"
Rims	2-ply 2x10 sides; front rim = flush beam
Beams	FRONT FLUSH (2)4x10 DF#1: 7'-9", 2 posts at 9" / 7'-3" (worst 1,821 lb)
Footings / posts	2 x 6x6 Douglas fir #1 with 4x4 knee braces on DP-50/50
Decking	TimberTech Reserve Sand — 24 rows parallel to the house, picture frame in Reserve Driftwood, 3/16" gaps, CONCEALoc
Rail	Fulton 36" Black: 5 bays, 6 posts, 31 LF
Stairs	none



Isometric from the stair corner



Isometric from the open corner

MATERIAL TAKE-OFF NET = counted off the drawings · ORDER = what to pull and ship

Lumber

NET	ORDER	UNIT	ITEM	WHY
13	**14**	ea	2x10x12 #1 True Frame Joist SYP GC	+1 cull (5 field joist · 2 PF joist · 2 left rim ply · 2 right rim ply · blocking, ledger · blocking, knee-brace block)
2	**2**	ea	4x10x8 Douglas Fir #1	exact (2 flush beam: FRONT FLUSH (2)4x10 DF#1)
2	**2**	ea	6x6x10 Douglas Fir #1	exact (yields 2 posts at ~8'-9 5/16")
2	**3**	ea	4x4x8 Douglas Fir #1 — knee braces	+1 (4 braces at 3'-9 15/16", 45° plumb cuts, 2'-6" legs: one along the beam, one up to the joists at every post)

Footings

NET	ORDER	UNIT	ITEM	WHY
2	**2**	ea	Diamond Pier DP-50/50 (head + 4 x 50" pins, plugs, caps)	exact — confirm stock
2	**2**	ea	Simpson ABA66Z adjustable post base 6x6 on the pier bolt	exact

Hardware

NET	ORDER	UNIT	ITEM	WHY
4	**5**	ea	1/2" x 10" HDG hex bolt + nut + 2 washers — knee braces (brace to the 6x6 post)	+1
2	**3**	ea	1/2" x 12" HDG hex bolt + nut + 2 washers — knee braces (brace to the (2)4x beam)	+1
2	**3**	ea	1/2" x 6" HDG hex bolt + nut + 2 washers — knee braces (brace to the joist block)	+1
14	**14**	ea	Simpson LUS28Z face-mount hanger 2x8/2x10	exact (7 joists x 2 ends)
4	**4**	ea	Simpson HUCQ210-2-SDS concealed-flange double 2x10 hanger (SDS screws included)	exact (2 side rims x 2 ends) — confirm stock
1	**1**	box	Camo 9ga HDG hanger nail 3" 5 lb (~350)	84 needed of ~350
1	**1**	box	Camo 9ga HDG hanger nail 1-1/2" 5 lb (~650)	56 needed of ~650
1	**1**	box	FastenMaster LedgerLOK 3-5/8" 50 ct	9 needed
4	**4**	ea	Simpson DTT1Z lateral tie kit w/ SDS screws	exact (2 near each end of every ledger, into house floor framing)
2	**2**	ea	Simpson CCQ column cap, 7" seat for the (2)4x10 beam on a 6x6 post (SDS screws incl.) — confirm SKU / engineer's equal	exact
10	**12**	ea	1/2" x 8" HDG carriage bolt + nut + washer	+2 ((2)4x beam plies through-bolted, 2 rows staggered @ 24" OC, 2" from top and bottom)
1	**1**	box	Simpson SD10212R100 #10 x 2-1/2" connector screw 100 ct	24 needed (bases 24 · caps 0)
1	**1**	box	GRK RSS #10 x 3-1/8" structural screw 100 ct	76 needed (laminations 52 · blocking 24)
12	**14**	ea	7/16" x 4-1/2" HDG hex bolt + nut + 2 washers	+2 (6 rail posts x 2)

Flashing & waterproofing

NET	ORDER	UNIT	ITEM	WHY
1	**1**	set	Ledger flashing set: SA membrane 6"x75' + (2) 10' alum. Z + end dams	exact
2	**2**	roll	Nichigo G-Tape 3035 2" x 65' black	93' needed of 130'
1	**1**	roll	Nichigo G-Tape 3035 4" x 65' black	30' needed of 65'

Decking

NET	ORDER	UNIT	ITEM	WHY
12	**14**	ea	TimberTech Reserve Driftwood 1x6x16 Grooved	+2 (A: 24 rows @ 7'-0", 2 per 16' board)
1	**1**	ea	TimberTech Reserve Driftwood 1x6x12 Square Edge — borders	exact (right border 11'-6 9/16")
1	**2**	ea	TimberTech Reserve Driftwood 1x6x20 Square Edge — borders	+1 (left border 11'-6 9/16" + front border 7'-11 1/8")

Fasteners

NET	ORDER	UNIT	ITEM	WHY
1	**1**	box	TimberTech CONCEALoc hidden fastener 500 ct (screws incl.)	168 needed · 500 supplied
1	**1**	box	Starborn Cap-Tor xd 2-3/4" Epoxy Driftwood 350 ct	106 screws needed (first row 7 · border 58 · fascia 41)

Fascia

NET	ORDER	UNIT	ITEM	WHY
3	**4**	ea	TimberTech Reserve Fascia Driftwood 0.58x11.95x12	+1 (left fascia 11'-6 1/16" · right fascia 11'-6 1/16" · front fascia 7'-10 3/16")

Rail

NET	ORDER	UNIT	ITEM	WHY
4	**4**	ea	TimberTech Fulton Rail 6' x 36" level panel Black	exact (cut to 5'-3 3/4")
1	**1**	ea	TimberTech Fulton Rail 8' x 36" level panel Black (pre-assembled steel balusters, cut to bay)	exact (cut to 7'-1 1/2")
2	**2**	ea	TimberTech Fulton 2" END post 36" Black w/ brackets, cap, skirt	exact
2	**2**	ea	TimberTech Fulton 2" LINE post 36" Black w/ brackets, cap, skirt	exact
1	**1**	ea	Rust-Oleum flat black touch-up (cut ends)	exact

Shade sail

NET	ORDER	UNIT	ITEM	WHY
1	**1**	ea	Shade sail — 4-corner shade sail, ~73 SF, Sand — SUPPLIED BY THE OWNER (GSX hangs and tensions it)	owner-supplied — not ordered

NET	ORDER	UNIT	ITEM	WHY
1	**1**	lot	Sail posts (2) — 4" steel, Black, the rail corner posts at the front of the deck	owner's price

FASTENER & CONNECTOR SCHEDULE

Knee braces	4x4 DF #1 at 45°, 2'-6" legs, both directions at every post: one along the beam toward mid-span, one toward the house up to a 2x10 block between the joists; (1) 1/2" through-bolt each end, 2" min. edge distance
Ledger	2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners
Beam	(2)4x plies through-bolted with 1/2" HDG bolts, 2 rows 2" from top and bottom, staggered @ 24" OC, a bolt pair 6" from each end; CCQ cap with SDS screws at each post
Joist tape	G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)
Field	CONCEALoc at every joist in every gap (168)
Face screws	Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then alternate top/bottom every 12", pre-drilled
Rail	Fulton 36" — 6 posts (2 CORNER, 2 END, 2 LINE) inside the outer rim ply, inner ply pocketed 2" wide, (2) 7/16" x 4-1/2" bolts per post

CUT LIST

MEMBER	STOCK	LENGTH	QTY	NOTE
Field joists	2x10	11'-1"	5	@ 16" OC, crown up
Picture-frame joists	2x10	11'-1"	2	centre 5 1/4" from each frame face
Side rim plies	2x10	11'-1"	4	2-ply, laminated on the ground with RSS 2 rows @ 12" staggered
Ledger	2x10	7'-9"	1	top at deck height less board thickness; membrane behind, flashing over
Blocking	2x10	1'-2 1/2"	6	1 row at 5'-8" from the wall
Field boards run A	Reserve Driftwood	7'-0"	24	boards run parallel to the house
FRONT FLUSH (2)4x10 DF#1 beam	4x10	7'-9"	2	y 11'-4"; posts at 9" / 7'-3"
Posts under FRONT FLUSH (2)4x10 DF#1	6x6	8'-9 5/16"	2	field-measure each
Knee braces	4x4	3'-9 15/16"	4	45° both ends, 2'-6" legs · along the beam toward mid-span + up to a block between the joists
Left border	Reserve Driftwood square	11'-6 9/16"	1	mitre at the corners; dividers on the doubled joist

MEMBER	STOCK	LENGTH	QTY	NOTE
Right border	Reserve Driftwood square	11'-6 9/16"	1	mitre at the corners; dividers on the doubled joist
Front border	Reserve Driftwood square	7'-11 1/8"	1	mitre at the corners; dividers on the doubled joist
Left fascia	fascia	11'-6 1/16"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia	fascia	11'-6 1/16"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia	fascia	7'-10 3/16"	1	flush under the board nose; full board at the front corners, short piece at the house

G-001 General notes · design criteria

G-001 GENERAL NOTES · DESIGN CRITERIA

Loads, codes, structure summary, sheet index, site — the permit-set cover

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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PROJECT

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech Reserve Driftwood, Fulton black, shade sail)
5549 S Lansing Ct, Englewood CO 80111
Prepared for Rachel Chiu
92 SF · 10'-0" above grade · 1 zone

DESIGN CRITERIA

Deck live load	40 psf uniform (IRC Table R301.5); guards 200 lb concentrated / 50 plf
Ground snow	30 psf (ground; deck design uses the larger of snow and live) — 40 psf live governs; 30 psf ground snow, roof 30 psf non-reducible (Arapahoe County design criteria); deck tables at 40 psf live + 10 psf dead
Dead load	10 psf deck
Roof snow (cover)	—
Design total	50 psf on the deck
Wind	115 mph Vult (3-second gust), Exposure B (suburban), Risk Cat. II — county design-criteria table (2015 cycle; confirm the 2021 table)
Frost depth	3'-0" (footings bear below)
Soil bearing	1,500 psf presumptive (IRC R401.4.1) — no soils report
Seismic	B
Wildfire (WUI)	not in a WUI zone
Codes	2021 IRC / IBC as adopted by Arapahoe County (Res. 21-394, eff. 4/1/2022); IRC R507 decks with the county-amended Colorado ICC Deck Building Guide; ASCE 7-16 loads; Tables R507.5 / R507.6 / R507.4 / R507.9
Engineering	Stamped structural set by a Colorado PE required — the stamped set governs

STRUCTURE

Footings	DP-50/50
Posts	2 x 6x6 Douglas fir #1 with 4x4 knee braces
Beams	FRONT FLUSH (2)4x10 DF#1: 7'-9", 2 posts at 9" / 7'-3" (worst 1,821 lb)
Joists	7 2x10 #1 True Frame Joist SYP GC @ 16" OC x 11'-1"
Rims	2-ply 2x10 sides; front rim = flush beam
Ledger	2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners
Decking	TimberTech Reserve Sand — 24 rows parallel to the house, picture frame in Reserve Driftwood, 3/16" gaps, CONCEALoc
Rail	Fulton 36" Black: 5 bays, 6 posts, 31 LF

SHEET INDEX

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S-102	Structure isometric, labeled
A-101	Decking plan
A-201	Railing plan + elevation
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A-301	Section + front elevation
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T-700	Order + schedules (takeoff)

SITE

Rear / front lot line: deck edge 13.0' from the line, setback per the PUD...
Side lot line: deck edge 23.7' from the line, 5' required
Easement: Check plat R1535678 for rear / side utility and drainage...
Unincorporated Arapahoe County (Cherry Creek CDP; Englewood mailing address). Parcel 2075-14-4-05-001 / PIN 031772427, Lot 41 Blk 9 Cherry Creek Vista 6th Flg Amend Resub (plat R1535678), 6,401 SF corner lot (front S Lansing Ct east, street side E Berry Ave north). R-P PUD Cherry Creek Vista (Z75-030 / P75-027 / P75-028). Deck on the WEST (rear) wall, north end, hung on the 2' second-floor cantilever that runs the full 33' wall (assessor sketch). Tax: CO 2.9 + RTD 1.0 + CD 0.1 + Arapahoe 0.25 = 4.25% (no city).

No scale

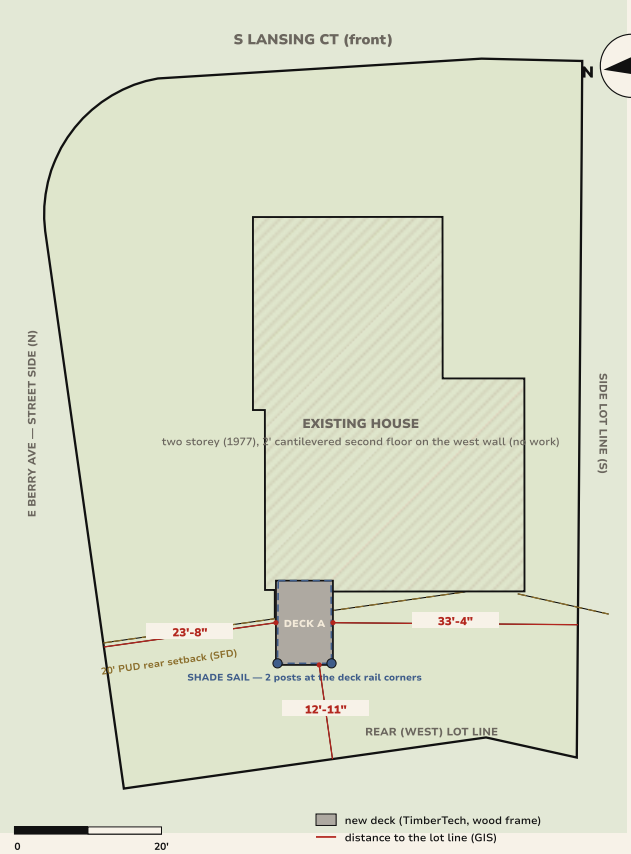
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

C-100 SITE PLAN

Parcel, existing house, the new deck, distances to the lot lines, setbacks

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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PARCEL

5549 S Lansing Ct, Englewood CO 80111
Parcel 2075-14-4-05-001 · PIN 031772427
Lot 41, Block 9, Cherry Creek Vista 6th Filing Amended Resub (plat R1535678)
Unincorporated Arapahoe County · zoning R-P PUD (Z75-030 / P75-027 / P75-028)
Lot 6,401 SF (GIS) · corner lot · flood zone X
House 1977, 2 storey, 1,704 SF + 660 SF basement + 441 SF garage (assessor)
HOA: Cherry Creek Vista II

LOT COVERAGE

Lot area	6,401 SF
Lot coverage limit	none in the PUD plan
Existing deck (assessor)	48 SF (8' x 6') — removed
New deck	8' x 12' · ~92 SF of frame, 10' above grade

SETBACKS

Elevated decks get no setback projection allowance (Arapahoe LDC) — the deck must meet the setbacks
GIS puts the new deck front ~13' off the west line: passes the 10' standard, not the 20' — confirm the governing column and the rear line (corner lot) with Zoning before submittal
E Berry Ave street side 10' required — ~23'-8" clear; south side 5' — ~33' clear
Measured (red): to the nearest lot line from the new work — GIS, not survey

SITE WORK

· Remove the existing 8' x 6' deck and its two posts / pads; haul off
· Two Diamond Pier DP-50 foundations (no excavation spoils), 6x6 DF #1 posts with 4x4 knee braces
· New ledger on the 2' cantilevered second floor per the engineer's detail; membrane + Z-flash, (4) DTT1Z lateral ties
· No stairs — the deck is reached from the house door only
· Call 811 before the piers; protect the lawn and the patio slab at the north wall

Arapahoe County GIS (address point, parcel, 2024 roof prints, R-P zoning, jurisdiction, flood, wildfire layers), county assessor record and sketch, 2nd Amended PDP P75-027 setback table, county deck permit guide (rev. 8/20/2024) and the county-amended ICC deck guide. GIS lines are not a survey — an ILC governs if Zoning asks.

Plan · approx. 1" = 20' at full size · GIS lines, not a survey

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

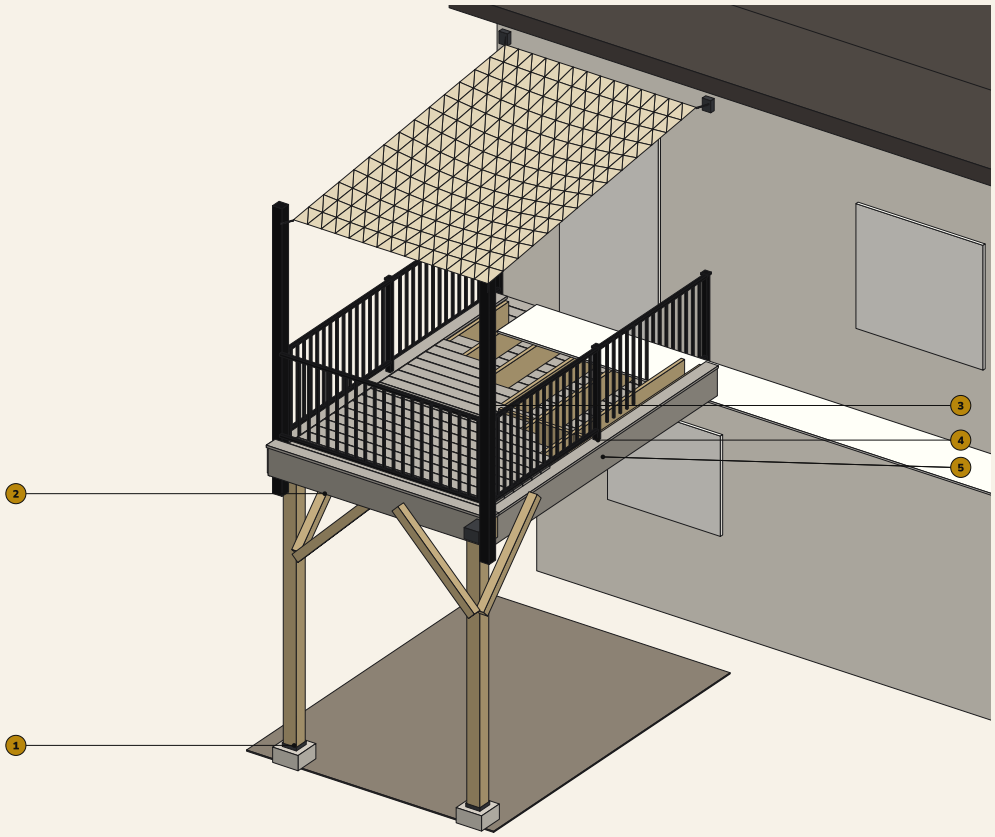
A-002 Isometric — assembled, labeled

A-002 ISOMETRIC — ASSEMBLED, LABELED

Everything you see called out: piers, decking, picture frame, skirt, stair, roof covers, finishes

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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KEY

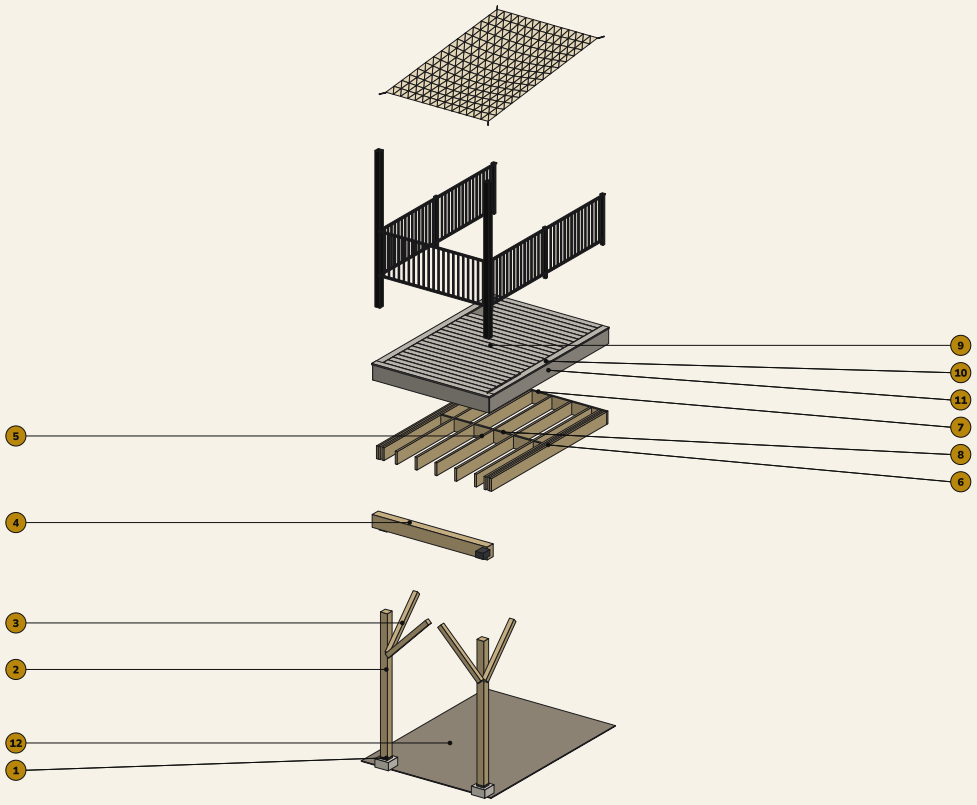
1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 4x4 DF #1 knee brace at 45°, both directions, through-bolted
3. TimberTech Reserve Driftwood field decking, hidden fasteners
4. Picture frame + breaker boards — TimberTech Reserve Driftwood
5. Fascia

House shown for context only (the deck work only). The frame under the decking is called out on S-102 and A-003.

Axonometric · not to scale · the model the plans are drawn from

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

A-003 Isometric — exploded, labeled



KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 6x6 Douglas fir #1 post on a ABA66Z standoff base
3. 4x4 DF #1 knee brace at 45°, both directions, through-bolted
4. Beam
5. 2x10 joist
6. Rim
7. Ledger
8. Blocking
9. TimberTech Reserve Driftwood field decking, hidden fasteners
10. Picture frame + breaker boards — TimberTech Reserve Driftwood
11. Fascia
12. Gravel over 6-mil poly under the decks

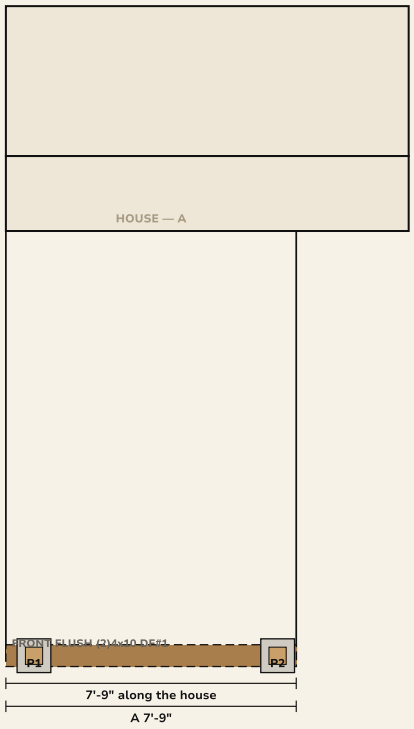
Layers lifted in build order. Lifts are for clarity only; every member returns straight down to its place.

S-100 FOUNDATION PLAN

Footings, posts and beam lines with the footing schedule; beams dashed (below the joists)

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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FOOTING SCHEDULE

Diamond Pier DP-50/50 (head + 4 x 50" pins), 3,300 lb allowable

ID	X	Y	ZONE	LOAD
P1	9"	11'-4"	A	1,821 lb
P2	7'-3"	11'-4"	A	1,821 lb

x from the left frame face facing the house; y from the reference wall.
Frost 3'-0" · soil 1,500 psf · worst post 1,821 lb

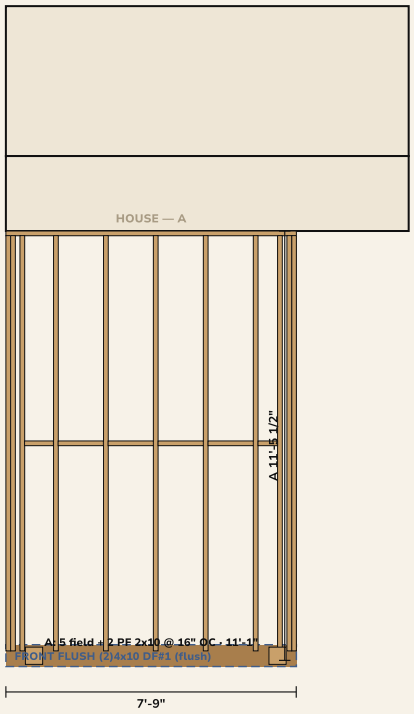
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

S-101 FRAMING PLAN

Ledger, joists, rims, blocking, beams; every member in the model — the cut list is on T-600

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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FRAMING NOTES

7 2x10 #1 True Frame Joist SYP GC @ 16" OC x 11'-1"

2-ply 2x10 sides; front rim = flush beam

2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners

blocking one row over the beam, staggered

G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

FRONT FLUSH (2)4x10 DF#1: 7'-9", 2 posts at 9" / 7'-3" (worst 1,821 lb)

Connectors:

1/2" x 10" HDG hex bolt + nut + 2 washers — knee braces (brace to the 6x6 post): 4

1/2" x 12" HDG hex bolt + nut + 2 washers — knee braces (brace to the (2)4x beam): 2

1/2" x 6" HDG hex bolt + nut + 2 washers — knee braces (brace to the joist block): 2

Simpson LUS28Z face-mount hanger 2x8/2x10: 14

Simpson HUCQ210-2-SDS concealed-flange double 2x10 hanger (SDS screws included): 4

Simpson DTT1Z lateral tie kit w/ SDS screws: 4

Simpson CCQ column cap, 7" seat for the (2)4x10 beam on a 6x6 post (SDS screws incl.) — confirm SKU / engineer's equal:

1/2" x 8" HDG carriage bolt + nut + washer: 10

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

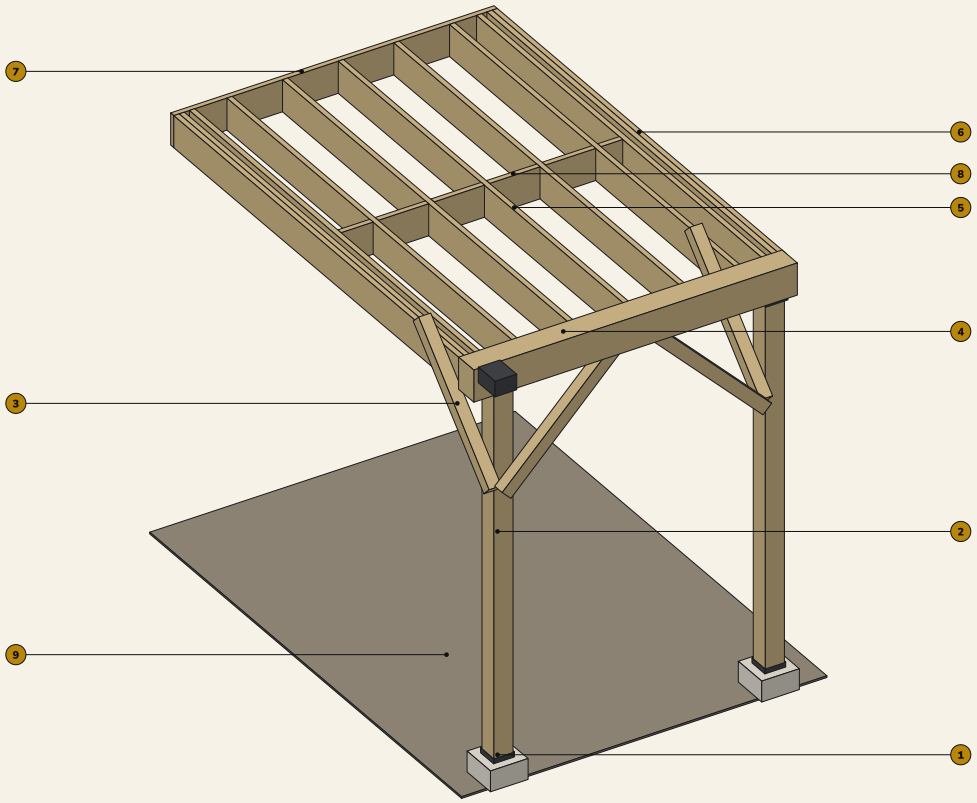
S-102 Structure isometric, labeled

S-102 **STRUCTURE ISOMETRIC — FRAME, LABELED**

Piers, post bases, posts, knee braces, post caps, beam, joists, double rims, ledger, blocking

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 6x6 Douglas fir #1 post on a ABA66Z standoff base
3. 4x4 DF #1 knee brace at 45°, both directions, through-bolted
4. Beam
5. 2x10 joist
6. Rim
7. Ledger
8. Blocking
9. Gravel over 6-mil poly under the decks

Decking, rail and house removed to show the frame. Joists hang from the ledger in face-mount hangers, bear on the drop beam with H2.5AZ ties; double rims in concealed-flange hangers.

Axonometric · not to scale · the model the plans are drawn from

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

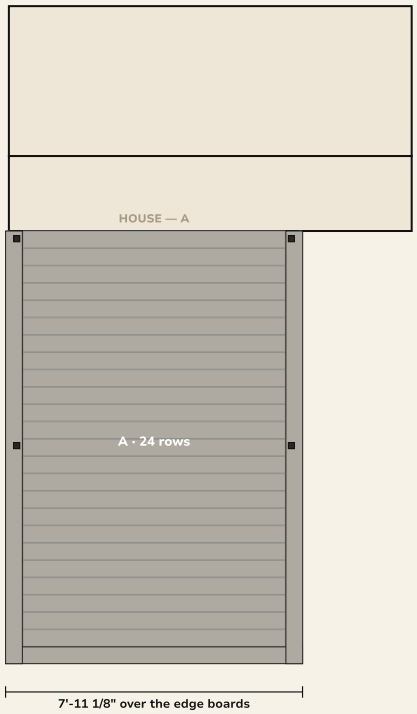
A-101 Decking plan

A-101 **DECKING PLAN**

Every board, border and divider as it lays; rail posts shown for the pockets

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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DECKING

Field: TimberTech Reserve Driftwood, 5.36" x 0.94", 3/16" gaps

Border: Reserve Driftwood

Boards run parallel to the house; edge overhang 1 1/16" over the fascia

CONCEALoc at every joist in every gap (168)

Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then

Rows per zone: A 24 (no rips)

Plan · not to scale · house at the top · dimensions govern

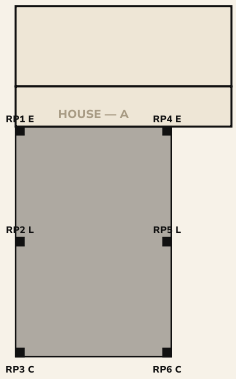
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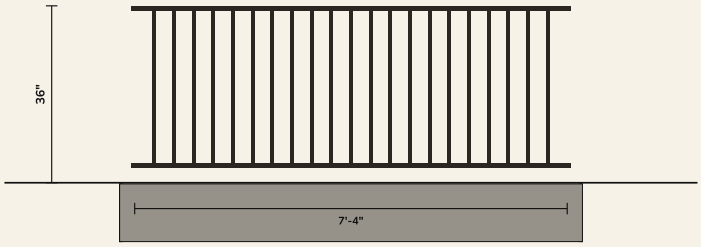
A-201 RAILING PLAN + ELEVATION

Post IDs and types (E end · L line · C corner), bay lengths, outer-edge elevation

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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RAILING

Fulton 36" Black: 5 bays, 6 posts, 31 LF

Fulton 36" — 6 posts (2 CORNER, 2 END, 2 LINE) inside the outer rim ply, inner ply pocketed 2" wide, (2) 7/16" x 4-1/2" bolts per post

Bays: left 5'-6 1/4" · left 5'-6 1/4" · right 5'-6 1/4" · right 5'-6 1/4" · front 7'-4"

Guard 36" min above the deck; openings under 4" (IRC R312). Rail LF for labor = deck-edge LF of actual rail.

Plan above · elevation of the outer edge below

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

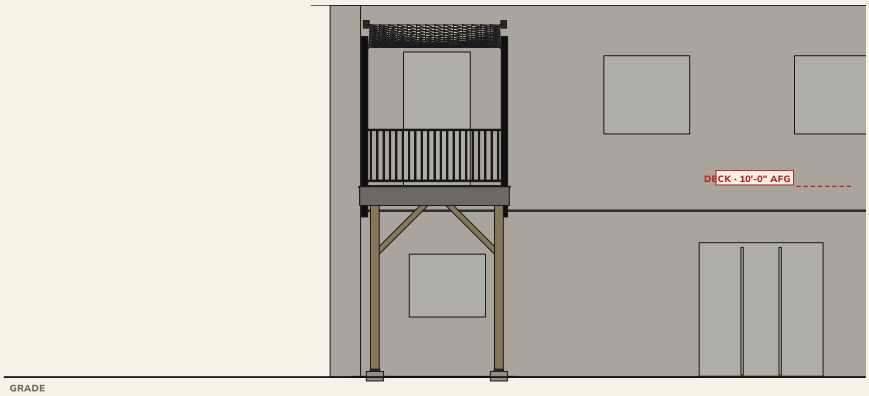
A-202 ELEVATIONS

Front from the yard and both ends, drawn from the model — deck and roof-cover heights

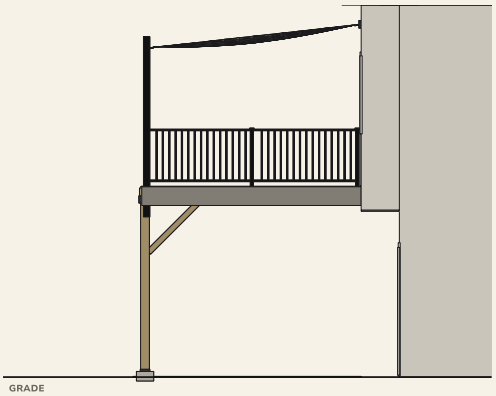
5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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FRONT ELEVATION — from the yard



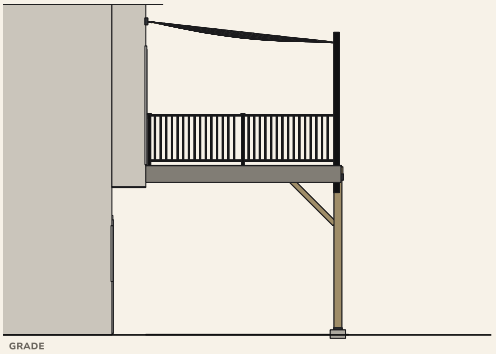
RIGHT END ELEVATION



NOTES

· Deck surface 10'-0" above grade
· Heights AFG = above finished grade at the beam line; field-verify grade before setting the piers
· Deck frame: 2x10 #1 True Frame GC @ 16" OC on a (2)4x10 DF #1 drop beam, 2 posts with 4x4 knee braces both ways
· Guard 36" min. on every open side — the surface is 10'-0" above grade (IRC R312.1)

LEFT END ELEVATION



Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

A-301 Section + front elevation

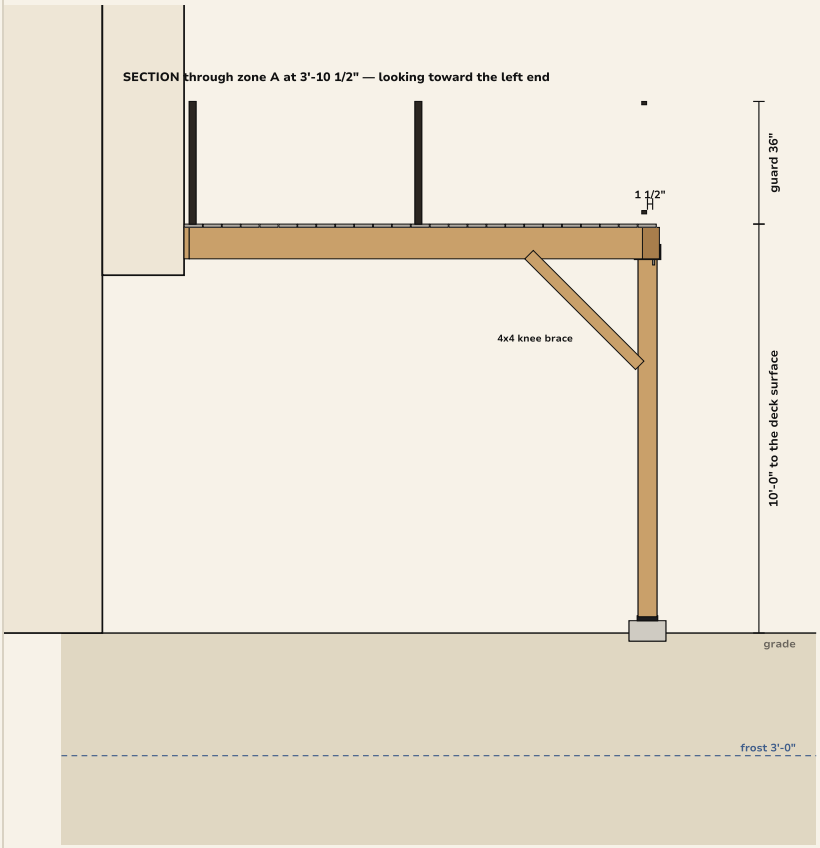
A-301 SECTION + FRONT ELEVATION

Cut through the deepest zone; heights to grade, frost line, cantilever, guard

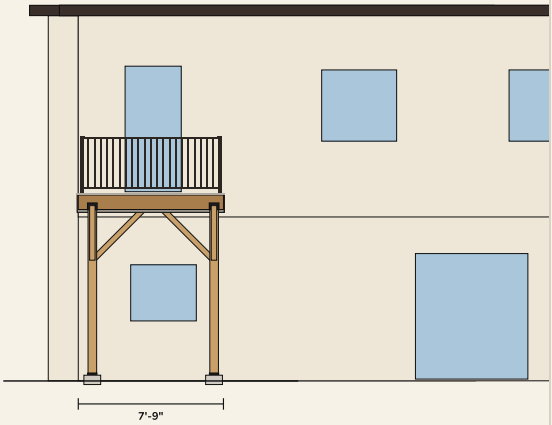
5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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SECTION through zone A at 3'-10 1/2" — looking toward the left end



FRONT ELEVATION — from the yard



SECTION NOTES

Deck surface 10'-0" above grade; guard 36" where the drop exceeds 30"

FRONT FLUSH (2)4x10 DF#1: 7'-9", 2 posts at 9" / 7'-3" (worst 1,821 lb)

Posts 2 x 6x6 Douglas fir #1 with 4x4 knee braces on DP-50/50

2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners

G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

Section left · elevation right · not to a common scale

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

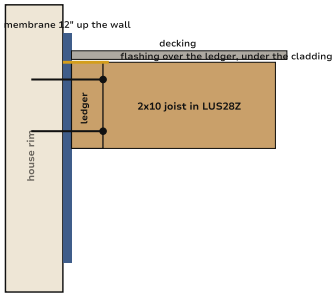
D-501 DETAILS

Ledger · post / beam / cap · footing · rail post · deck edge · the job's special condition

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

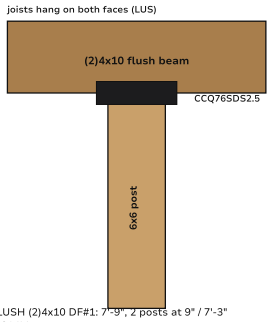
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1 LEDGER AT THE HOUSE



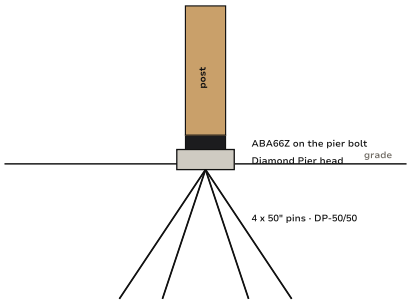
2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners

2 POST · BEAM · CAP

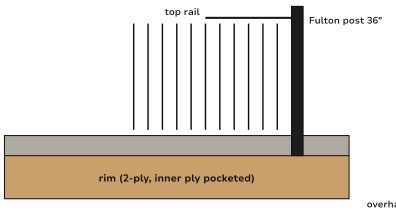


FRONT FLUSH (2)4x10 DF#1: 7'-9", 2 posts at 9" / 7'-3" (worst 1,821 lb)

3 FOOTING · POST BASE

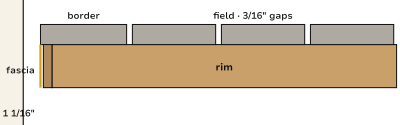


4 RAIL POST · GUARD



Fulton 36" — 6 posts (2 CORNER, 2 END, 2 LINE) inside the outer rim ply, inner ply pocketed 2" wide, (2) 7/16" x 4-1/2" bolts per post

5 DECK EDGE · BORDER



CONCEALoc at every joist in every gap (168)

6 FLUSH BEAM

Flush beam: top of beam = top of joists. Joists hang on both faces in LUS hangers; opposing hangers per Simpson and the stamped set.

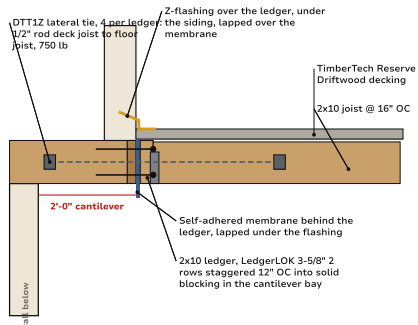
D-502 CONNECTION DETAILS — FRAME

Ledger on the cantilever · hangers · joists on the beam · (2)4x10 on the cap · pier + base · knee braces

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

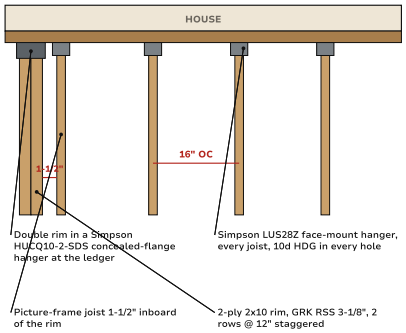
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1 LEDGER ON THE CANTILEVERED FLOOR — SECTION

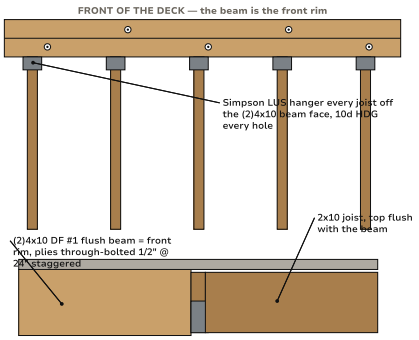


ENGINEER: Arapahoe County bars a ledger on a cantilevered floor without engineer approval. The stamped detail governs: ledger into the blocked cantilever joists, or a freestanding beam on posts at the house.

2 HANGERS — PLAN AT THE LEDGER + SIDE RIM

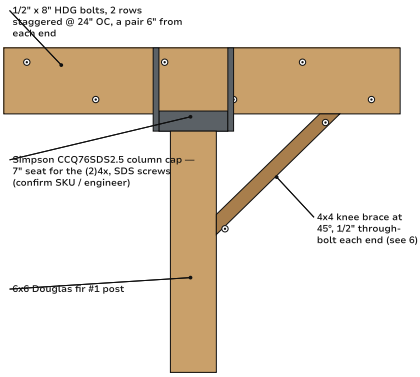


3 JOISTS INTO THE FLUSH BEAM — PLAN + SECTION

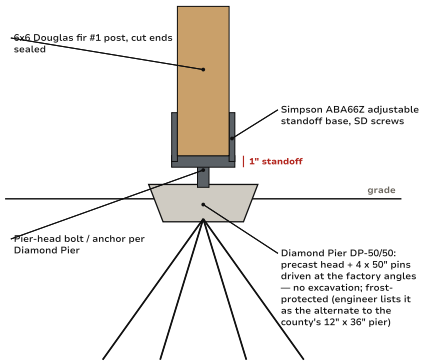


No cantilever: the beam carries the front edge, joists hang both ends

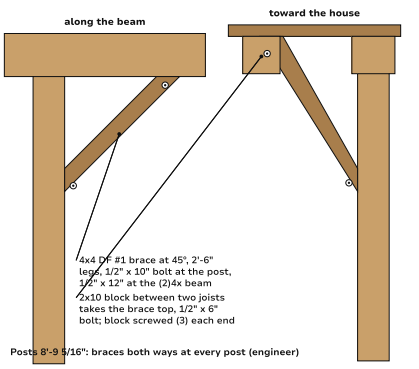
4 (2)4x10 BEAM ON THE 6x6 — ELEVATION



5 POST BASE ON THE DIAMOND PIER — SECTION



6 KNEE BRACES — BOTH DIRECTIONS



Posts 8'-9 5/16": braces both ways at every post (engineer)

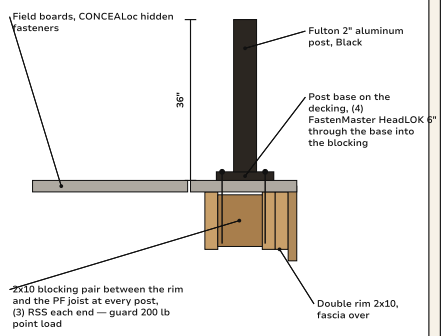
D-504 CONNECTION DETAILS — DECK EDGE, RAIL, LATERAL

Rail post mounting · picture frame + fascia · guard bay · DTT1Z lateral ties · joist tape · fastening schedule

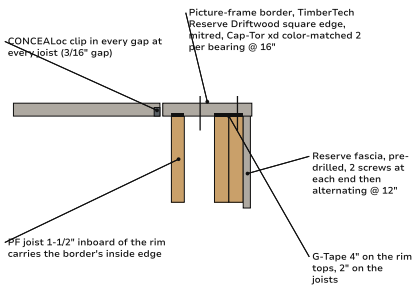
5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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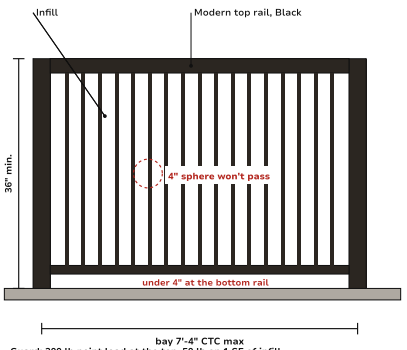
1 RAIL POST AT THE RIM



2 DECK EDGE — PICTURE FRAME + FASCIA

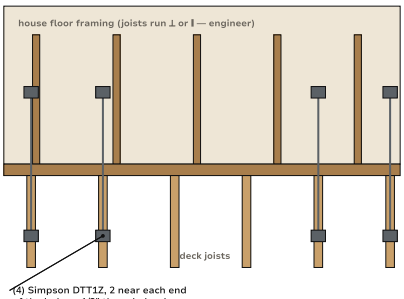


3 GUARD — ONE BAY, ELEVATION



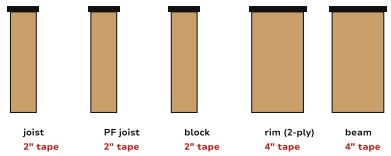
Guard: 200 lb point load at the top, 50 lb on 1 SF of infill (IRC Table R301.5) — submit the rail code report

4 LATERAL LOAD — DTT1Z TIES (PLAN)



(4) Simpson DTT1Z, 2 near each end of the ledger: 1/2" threaded rod through the ledger and the band into a floor joist (or blocking) inside — 750 lb each (IRC R507.9.2)

5 JOIST TAPE + BLOCKING

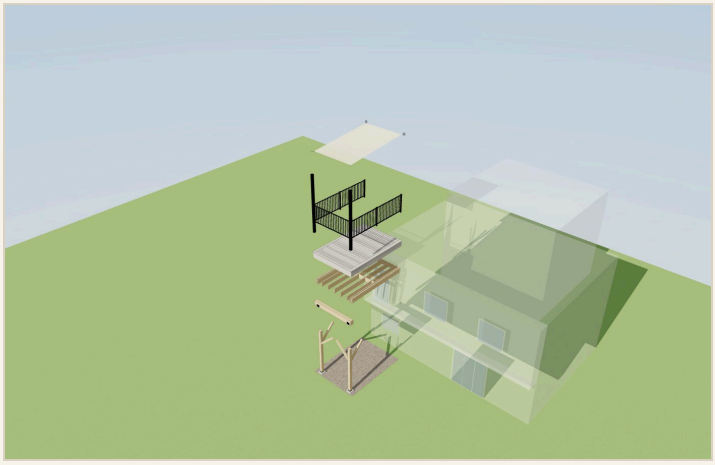


Nichigo G-Tape 3035 (black) on every top edge before decking, lapped over the ends and down the sides 1/2"; one row of 2x10 blocking over the beam, (3) structural screws each

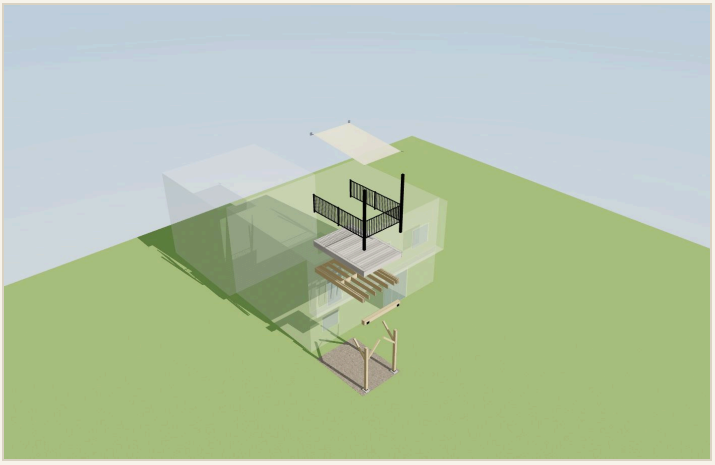
6 FASTENING SCHEDULE

Ledger to house	LedgerLOK 3-5/8", 2 rows staggered 12" OC — engineer
Lateral	(4) DTT1Z, 1/2" rod into the floor framing
Joist to ledger	LUS28Z, 10d HDG every hole
Rim to ledger	HUCQ10-2-SDS, SDS screws
Rim plies	GRK RSS 3-1/8", 2 rows @ 12" staggered
Joist to beam	H2.5AZ each joist + rim, 10d x 1-1/2"
Beam plies	1/2" x 8" HDG bolts, 2 rows staggered @ 24"
Beam to post	CCQ76SDS2.5 cap, SDS screws
Post to pier	ABA66Z on the Diamond Pier, SD screws
Knee braces	1/2" through-bolt each end (x10 / x12 / x6)
Field boards	CONCEALoc, every joist, every gap
Borders / fascia	Cap-Tor xd color-matched, pre-drilled
Rail posts	(2) 7/16" bolts per post

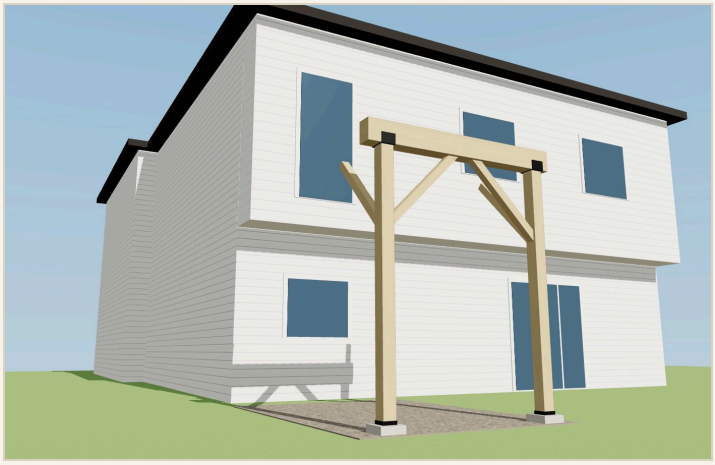
EXPLODED VIEWS



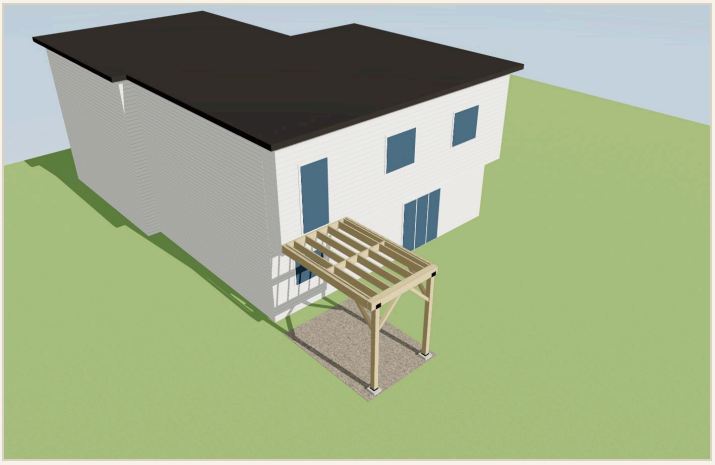
Exploded — every layer



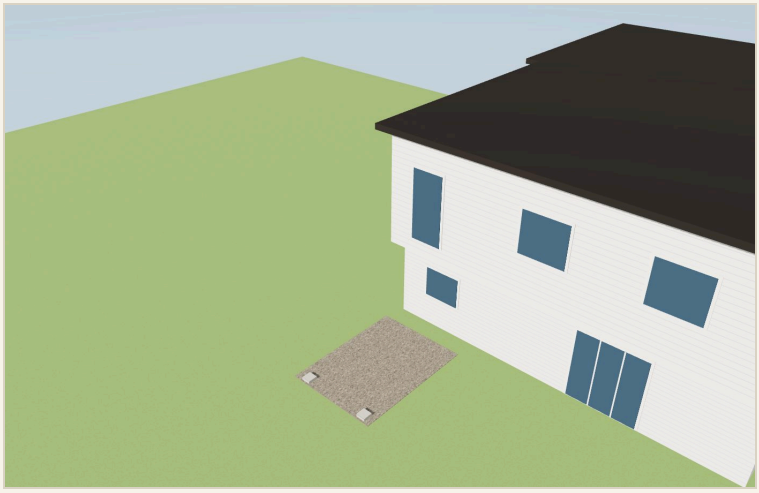
Exploded — from the open corner



Posts and beams on the footings



Frame only — joists, rims, beams, ledger



Step 1 · Footings — 2 Diamond Pier DP-50/50 driven-pin footings

T-400 · Step 1 · Footings

GOES IN

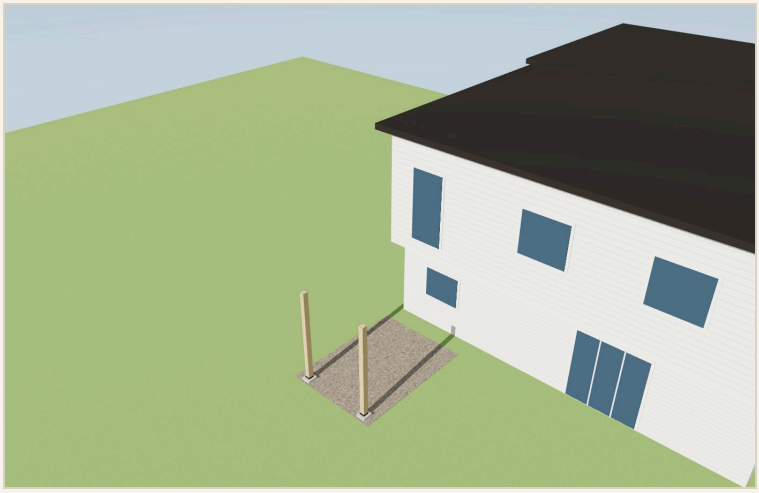
- (2) Diamond Pier DP-50/50 (head + 4 x 50" pins, plugs, caps)
- (2) Simpson ABA66Z adjustable post base 6x6 on the pier bolt

HOW

- Call 811 and locate before any hole or pin. Strip the existing deck first.
- Pull the beam lines from the house wall: FRONT FLUSH (2)4x10 DF#1 at 11'-4" from the reference wall, posts at 9" / 7'-3".
- Set 2 Diamond Pier heads level at grade on undisturbed soil and drive the 4 pins in each; caps on.

CHECK

- Footings on the layout within 1/2"; bearing on undisturbed soil
- Pins fully driven, heads level
- Photos before backfill



Step 2 · Posts — 6x6 Douglas fir #1 on ZMAX bases

T-401 · Step 2 · Posts

GOES IN

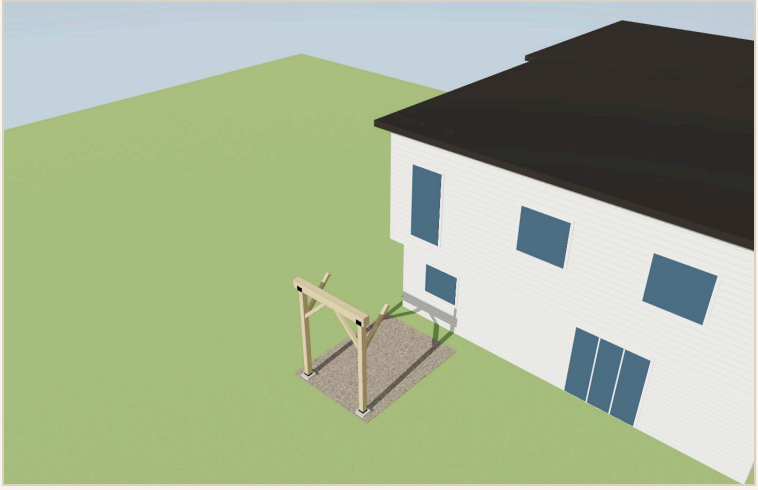
- (2) 6x6x10 Douglas Fir #1
- (2) Simpson ABA66Z adjustable post base 6x6 on the pier bolt

HOW

- 6x6 posts on the bases; cut so the beam top lands at the joist bottom (drop beam) or the joist top (flush beam): FRONT FLUSH (2)4x10 DF#1 posts 8'-9 5/16" — field-measure each.
- Plumb both ways, brace, then the caps go on with the beams.

CHECK

- Posts plumb to a string; tops on the beam line
- No post in ground contact without a standoff



Step 4 · Beams — FRONT FLUSH (2)4x10 DF#1 (2 posts)

T-403 · Step 4 · Beams

GOES IN

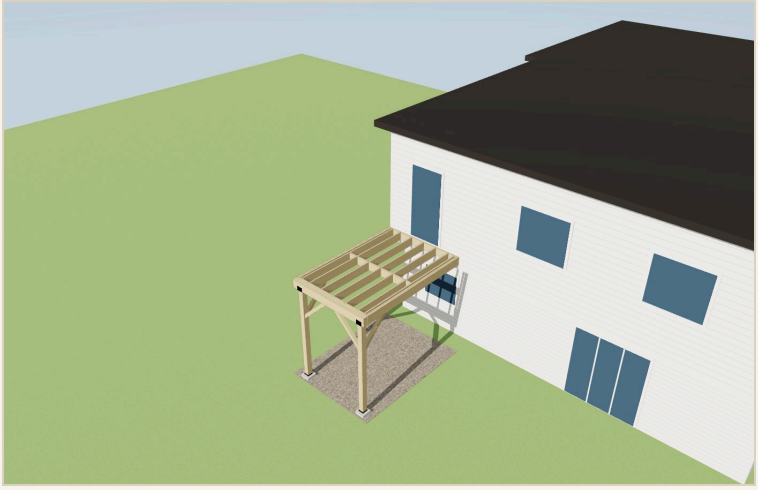
- (2) 4x10x8 Douglas Fir #1
- (2) 6x6x10 Douglas Fir #1
- (2) Simpson CCQ column cap, 7" seat for the (2)4x10 beam on a 6x6 post (SDS screws incl.) — confirm SKU / engineer's equal
- (2) 4x4x8 Douglas Fir #1 — knee braces
- (4) 1/2" x 10" HDG hex bolt + nut + 2 washers — knee braces (brace to the 6x6 post)
- (2) 1/2" x 12" HDG hex bolt + nut + 2 washers — knee braces (brace to the (2)4x beam)
- (2) 1/2" x 6" HDG hex bolt + nut + 2 washers — knee braces (brace to the joist block)
- (10) 1/2" x 8" HDG carriage bolt + nut + washer

HOW

- FRONT FLUSH (2)4x10 DF#1: 7'-9" one piece, 2 posts @ 6'-6" max
- Crown up, strung straight, caps screwed off with the SDS screws in the box.
- Two-ply beam: through-bolt the plies with 1/2" HDG bolts, 2 rows 2" from top and bottom, staggered @ 24" OC, a pair 6" from each end — before it goes up.
- Knee braces once the beam is capped: 4x4 at 45°, 2'-6" legs, one along the beam toward mid-span and one toward the house up to a 2x10 block between two joists (set the block with the frame) — 1/2" through-bolt each end, then pull the temporary bracing.

CHECK

- Beam top = joist bottom (drop) / joist top (flush)
- Splices land on a post; caps fully fastened
- Every knee brace bolted both ends; posts still plumb



Step 5 · Frame — 2x10 joists at 16", rims, ledger, blocking over the beam

T-404 · Step 5 · Frame

GOES IN

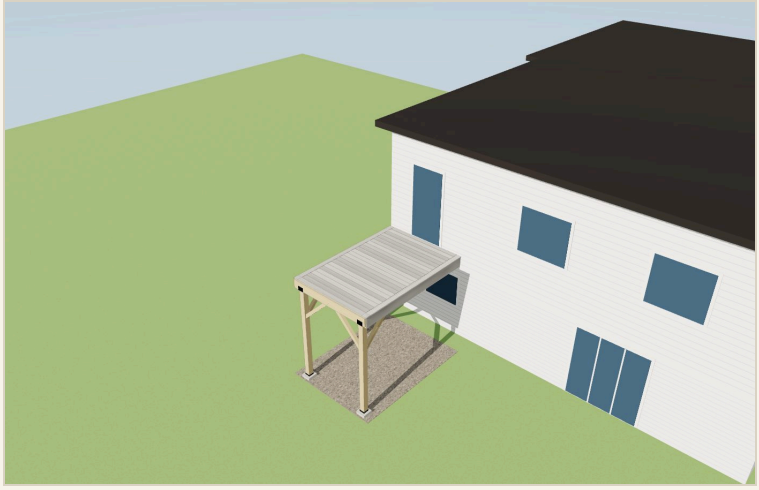
- (13) 2x10x12 #1 True Frame Joist SYP GC
- (14) Simpson LUS28Z face-mount hanger 2x8/2x10
- (4) Simpson HUCQ210-2-SDS concealed-flange double 2x10 hanger (SDS screws included)
- (1) Camo 9ga HDG hanger nail 3" 5 lb (~350)
- (1) Camo 9ga HDG hanger nail 1-1/2" 5 lb (~650)
- (1) FastenMaster LedgerLOK 3-5/8" 50 ct
- (1) Ledger flashing set: SA membrane 6"x75' + (2) 10' alum. Z + end dams
- (2) Nichigo G-Tape 3035 2" x 65' black
- (1) Nichigo G-Tape 3035 4" x 65' black

HOW

- 2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners
- A: 5 joists @ 16" OC x 11'-1". Crown up, hangers at the ledger, both faces of the flush beam.
- Blocking one row over the beam, staggered.
- G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

CHECK

- Frame square: diagonals within 1/8"
- Every hanger fully nailed; ledger fasteners per the schedule
- Tape continuous; photos of the ledger, hangers and footings before decking



Step 6 · Decking — TimberTech Reserve Driftwood, hidden fasteners

T-405 · Step 6 · Decking

GOES IN

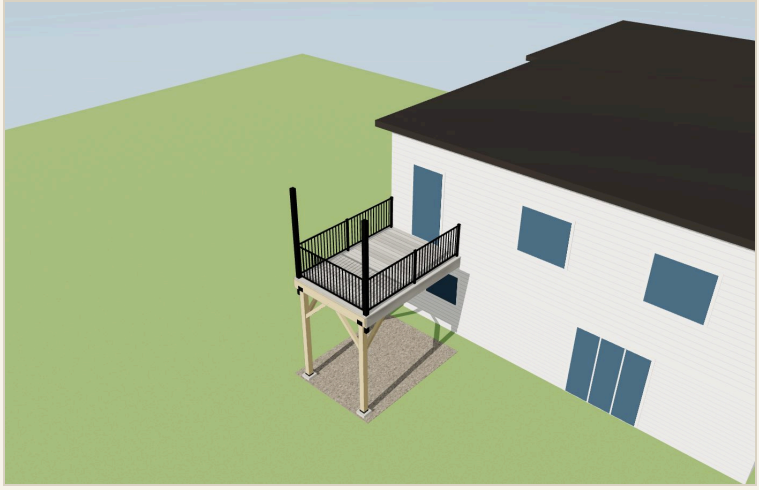
- (12) TimberTech Reserve Driftwood 1x6x16 Grooved
- (1) TimberTech Reserve Driftwood 1x6x12 Square Edge — borders
- (1) TimberTech Reserve Driftwood 1x6x20 Square Edge — borders
- (3) TimberTech Reserve Fascia Driftwood 0.58x11.95x12
- (1) TimberTech CONCEALoc hidden fastener 500 ct (screws incl.)
- (1) Starborn Cap-Tor xd 2-3/4" Epoxy Driftwood 350 ct

HOW

- CONCEALoc at every joist in every gap (168)
- Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then alternate top/bottom every 12", pre-drilled
- Rows per zone: A 24. Borders and breakers first, then the field rows from the outer edge toward the house.

CHECK

- 3/16" gaps held
- No cut end shows at the outer edge
- Color: every board from the same run



Step 7 · Rail — Fulton 36" Black

T-406 · Step 7 · Rail

GOES IN

- (4) TimberTech Fulton Rail 6' x 36" level panel Black
- (1) TimberTech Fulton Rail 8' x 36" level panel Black (pre-assembled steel balusters, cut to bay)
- (2) TimberTech Fulton 2" END post 36" Black w/ brackets, cap, skirt
- (2) TimberTech Fulton 2" LINE post 36" Black w/ brackets, cap, skirt
- (1) Rust-Oleum flat black touch-up (cut ends)

HOW

- Fulton 36" — 6 posts (2 CORNER, 2 END, 2 LINE) inside the outer rim ply, inner ply pocketed 2" wide, (2) 7/16" x 4-1/2" bolts per post

CHECK

- Top of rail 36" above the decking; posts plumb
- Cables tensioned / balusters under 4" clear
- End of construction

3D MODEL



The interactive 3D model lives at the online version of this document.

Quantities from the drawings; the stamped engineering set governs structure. Quote requested — no pricing on this sheet.